PROJECT CHARTER
For REDESIGN3D

Reimagine the art of business cards. Get them redesigned, customized and be the first to stand out.

Prepared for:
REDESIGN3D
New Jersey Institute of Technology
Newark, NJ, 07102
Attn: Prof. Dipesh Patel
dipesh.patel@njit.edu

Prepared By:
Vitaliy Gotsko
Daniel Ivanov
Maximilian Vitalea
Sean Romero
REDESIGN3D
New Jersey Institute of Technology
Newark, NJ, 07102
redesign3d@zoho.com
9734159291

Contents

Executive Summary 2
Project Purpose 2
Business Objectives 2
Project Details 2
Requirements 2
Deliverables 2
Estimated Schedule 2
Estimated Budget 3
Human Resource Requirements 3
Procurements 3
Completion Criteria 3
Define Project Success 4
Alternatives 4
Stakeholders 4
Project Manager 4
Project Authorization ***(For Professor Use ONLY)*** 4

Executive Summary

Provide high-level, summary information about the project and why it is needed.

This project is about raising business card standards and to get people more noticed in the process. Using refined plastic filament and state of the art CAD software and 3D printers it's possible to create the most unique and memorable business cards. With addons like near field communication technology (NFC), the business cards can go beyond just being a bunch of text on a rectangle, they can contain any link, phone number, blog or website, and become the introduction of your empire no matter how big or small your business is or if it's personal or corporate. The purpose of this innovation is to redefine what it means to introduce yourself to others with the measures of business cards. Standing out was never more easy. Being highly customizable with the use of modern CAD software like Solidworks there is nothing standing in the way of being the most impressive one in the room. This is not just about business cards, it's about speed, ease of use, price, and expert advertising. With a timeline of two months, this project aims to finalize after acquiring and using the proper materials like filament, packages for the product, marketing tools and figuring out distribution methods. Final goals would be to make the same, high quality product cheaper by perfecting consistency of outcome and minimize machine failure.

Project Purpose

With greater specificity, state the purpose of the project:

The purpose of this project is to design, develop and implement a line of highly customizable 3D printed business cards that have aesthetic designs. Traditional business cards often fail to leave a lasting impression and the 3D printing offers an opportunity to create a durable and visually appealing business card that reflects the owner's brand identity. This project supports the growing demand for sustainable business solutions by using eco-friendly materials. There are many ways to express and show yourself to the world like dressing a certain way, driving a certain car, or even the way you post online. When it comes to business, it's best to be concise, practical and with a hint of style, which is why business cards exist. That little rectangle allows for clear and quick communication between parties without wasting extra space or time. This method allows for everyone to have enough information to decide for themselves what they want to do with it. This is why the presentation of your business card must be premium. Designed to perfection from corner to corner, with the most precise equipment on the market. Most business cards are paper, made with basic templates without any imagination or effort put behind them, and end up being thrown out a minute after or stuffed deep into suit pockets until they eventually fall out or disintegrate from being at the dry washers one too many times. Using refined plastic filament and state of the art CAD software and 3D printers it's possible to create the most unique and memorable business cards. They do not break easy and leave quite the impression. Our goal is to get you noticed, and with our team and resources it won't be hard. From the initial stages of design to the manufacturing of your best image that we call the canvas of opportunity made by yours truly. Not only that but customers can pitch in with the design or even suggest a full scope with which we fully stand behind. Our customers have a variety of choices, requests, and opportunities for business.

Business Objectives

A project should support the objectives in the strategic plan. List specific strategic plan objectives and describe how the project supports each one.

- Increase inflow of customers
  - Establish and run a functional E-commerce website with simple and easy to understand instructions on how to order our products.
- First month sales of 100+ units after launch
  - Starting out as word of mouth, before official marketing launch, cold call and email businesses preaching our message.
- Increase customer engagement
  - By adding new features like NFC add ons, users will be able to engage in business card creation. Allow customers to have choices so we see what they tend to like more through their click paths.
- Reduce production speed and cost
  - Using different 3D printer settings and modules, without compromising quality. Purchase more expensive equipment which is able to use multiple colors in a print more effectively.
- Expand our brand throughout the world
  - Using social media apps to spread our business through social media. Which could attract new customers and business. Demonstrating that we are not your average E-commerce business. We wish to expand our horizons and grow, and help others grow too. By using apps like TikTok and Instagram. To reach a bigger audience.

Project Details

Describe the project of choice in greater detail.

This project is about displaying, creating, and delivering quality business cards to customers. The process would first start with a person noticing an advertisement or hearing about our company website. After that, they visit our website and choose a business card that best suits them, either that or they call us and consult the process with their vision. From the list of other options, they either add or ignore them and check out. We calculate the execution time for the product and give them the estimated arrival date. We take the input into our software and complete the design, which after depending on the amount of cards needed, the time of printing them all is calculated based on current hardware limitations. Shipping is handled using FedEX or other cheaper alternatives and even personal deliveries by one of our own.

Requirements

List the high-level project requirements.

Some project requirements would be:

1. Design Software access: Using our student status, and free CAD software online like the Bambu Slicer and TinkerCAD, as well as CuteCutter for SVG files and even microsoft paint.
2. 3D printing equipment: We need to purchase 3D printers and learn how to use them properly by design as well as maintain them. Such products would include the Bambu Lab A1 and A1 mini for affordability.
3. NFC technology: purchase NFC stickers on Amazon and download NFC editable software.
4. Website: Create a website using a domain name, either by paying a service like GoDaddy or similar.

5. Materials: Purchase PLA filament and tools to handle 3D printing tasks from Amazon and Microcenter.
6. Marketing: Deploy team members to talk about our project and E-Commerce business to get the word flowing. Have tabling events for free at colleges by helping out clubs.

Deliverables

List the project deliverables. A deliverable is a unique and verifiable product, result or capability to perform a service that must be produced to complete a process, phase or project.

- *Prototype Testing* Phase
  - 3D Model a set of prototype 3D printed business cards that demonstrate a variety of interesting designs. After the models are made, the cards will be printed utilizing a 0.2mm nozzle paired with Bambu Lab's state of the art A1/A1 mini printer to ensure fine quality. Additionally, two of the demonstration cards must contain a sample NFC add-on and a QR code. The prototypes will exhibit different design patterns, colors, aesthetics, and technologies.
- *Functional E-Commerce Website* Phase
- A fully operational e-commerce website that allows customers to browse and order sample 3D business cards. Customers will have the ability to "customize" one of our designs with text, font, and colors of their choosing. The website will include product galleries, pricing information, and checkout options.
- *Marketing Funnel & Branding Materials* Phase
  - Establish a brand identity to accompany the website launch. Deliverables will include a logo, color scheme, typography guidelines, and digital banner(s).
  - Create social media accounts on platforms like Instagram, TikTok, and LinkedIn to increase brand visibility, share product updates, and as funnels to attract new customers.
- *Logistics*
  - Tracking System
    - This must be either simply spreadsheet based or integrated into our website backend. A tracking system will provide us the ability to effectively track inventory levels, job costing, job progress, and shipping status for every batch/orders of cards produced. This system will serve as inventory tracking and as a job tracking system. Adding on to the benefits of job tracking, all logistics updates and documentation will be compiled to verify every customer receives their product in a timely manner. This will ensure that every material is costed and tracked, a timeline for a job is logged and updated, and customer's orders will arrive at their doorstep.

Estimated Schedule

Provide high-level schedule information.

Project Milestones and/or Phases	Estimated Completion Date
Start of Project	Oct 17
Concept Design	Oct 22
Prototype Development and 3D Printing	Nov 9
Minimally-Functional Website Launch	Nov 20
Branding Identity	Nov 20
Marketing Campaign Kickoff	Dec 1
Product Launch	Dec 7

Estimated Budget

Provide high-level budget information.

Resource Description	Estimated Cost
3D Printers: 3x A1 Minis with Automatic Material System (AMS)	$1200
Website Development / Maintenance	$200/yr
Website Editor: if with GoDaddy	$120/yr
Marketing & Advertisement	Free
Packaging & Shipping Supplies	$100
Filament: 10 Rolls of different colors	$250
Electricity	$45/month
Business Registration	$125
TOTAL	**~$2,535**

Human Resource Requirements

List the people from within the organization that will be assigned to the project.

Name	Department/Title	Contact Information	Immediate Supervisor

Vitaliy Gotsko	3D Modeling/ Manufacturing	vg534@njit.edu	
Daniel Ivanov	Technology / Manufacturing	dmi@njit.edu	
Maximilian Vitale	Marketing	mjv55@njit.edu	
Sean Romero	Marketing	slr36@njit.edu	

Procurements

List the known resources which must be procured.

Description	Source / Vendor	Estimated Cost Subtotal (per)
3D Printers	Bambu Store, Microcenter, BestBuy	$399.00
Flat / Textured Build Plates	AliExpress, Amazon	$5.00 - 10.00
PLA Filament Spools	Microcenter	$18.99/kg
NFC Tags (50 pcs)	Amazon	$15.00
Plastic Glue	Amazon	$7.99
Order Packaging (White Literature Mailers)	ULINE	$0.65/EA Qty 50 = $32.50

Completion Criteria

What must occur before the project is considered complete?

- Prototype testing is completed.
  - All prototype 3D business cards, including both NFC-enabled and QR code variations, must be designed and printed. The set of prototype business cards will encompass our product offering fully. We will have cards of different colors, texts, and fonts and cards that show our technologies such as our embedded NFC transmitter technology.
  - Printing parameters for each card are documented and calculated so the team will have quantifiable data of individual card cost, elapsed printing time, and rates.
- Website Fully Operational
  - The e-commerce website must be live, functional, and *at least* capable of processing real customer orders. The order process will include design options, design customization, checkout, and confirmation of features.
- Branding & Marketing Launched
  - Create a broad Redesign3D brand identity schema .pdf file. The pdf file will include a version(s) of the Redesign3D logo, such as a long form and condensed version. With that, blocks of our individual brand colors with their respective hexadecimal codes. Text with our typography that is used in our branding, marketing, and website.
  - Social media accounts through social media platforms like Instagram (main page), Tiktok, and LinkedIn. A couple of posts will be deployed with some launch content and with links to our website for customer leads.

- Logistics System Implemented
    - A working logistics system either implemented through a simple spreadsheet based system or integrated with our website's backend. The system will accurately track materials, print jobs, and shipping progress.
    - Material inventory (filament details (color, manufacturer, tracked weight), NFC tags, packaging) must be logged.
    - Each order must be documented from order start to order delivery.

Define Project Success

What specific measurable objectives must the project achieve to be considered successful?

- Sale Target:
    - Achieve at least 150 projects sold. Showing our dedication to sending out projects fast and looking good.
- Website Performance:
    - The E-commerce website must work properly and be able to attract at least a good number of customers who interact with our website.
- Social media engagement:
    - A working social media platform showcasing our unique work. Also, having about 150 followersto show how real we are and how much we want to grow.
- Feedback from Customers:
    - Get 15 customer reviews showing our work is nothing but unique, and put it up on our website. As well as showing that we are a real business, trying to make life easier while also trying to make things interesting.
- Logistics:
    - Achieve a system that will update the customers about their orders. While making sure all orders can get to their respective destination on time.
- Reduction for materials:
    - Reduce the need for buying many PLA by recycling unused PLA to create new filament. Reducing the price needed for more filament each time. At least reduce the cost of materials needed since they are expensive and will ruin our profit.

Alternatives

List the known alternatives to undertaking the project and state the pros and cons of each.

Other alternative ways that our company could put its resources into and pursue are:

1. Prototyping: we could prototype designs our clients give us. A pro of that would be higher payout, whereas a Con would be the need for an employee with strong and experienced CAD.
    a. Our current choice is better because although this version may have a higher status and payout, business cards will always be in style and we are able to keep up with the technological advancements too.

2. Trendy toy manufacturing: Creating toys that are trending at the moment to profit off of temporary hype. The pro's would be that the designs are usually already out there and are free whereas cons would be reliant on how well we advertise better than other companies that do that already as well as relying on the trends to stick around.
    a. Our choice is better than this one because toy manufacturing is strong but it comes with risks of poor quality when designs are complex unlike business cards that have a simple flat design and are the same size every time.
3. Produce regular business cards with basic features and no fancy add ons or graphics: Pro's would be ease of design and not having to worry about making errors and receiving complaints as well as having an easier time to pass quality checks consistently and cut costs. Con's would be losing out on customers who want to be more expressive and stand out.
    a. This is a good business but we just chose the enhanced version of it to apply to more customers.
4. 3D manufacturing 3D models: Simply print out 3D models people send us and charge them for the materials and service fee plus shipping. Pro's would be that we wouldn't require to design anything ourselves, Cons would be the competition with other companies like Xometry who may do it for cheaper or have better quality and customer service.
    a. This is a good business but it comes at the expense of precision and quality control of the 3D models. People want all kinds of things 3D printed and things can be very complicated to execute. Poor execution leads to poor reviews. Making business cards is consistent and won't require a learning curve as with the adjustments you'd have to make to make sure a difficult print doesn't fail.

STAKEHOLDERS

List all known project stakeholders.

Name	Project Role	Organization	Contact Information
Vitaliy Gotsko	3D Modeling/ Manufacturing	REDESIGN3D	vg534@njit.edu
Daniel Ivanov	Technology / Manufacturing	REDESIGN3D	dmi@njit.edu
Maximilian Vitale	Marketing	REDESIGN3D	mjv55@njit.edu
Sean Romero	Marketing	REDESIGN3D	slr36@njit.edu
Prof. Dipesh Patel	Professor / Investor	NJIT	dipesh.patel@njit.edu
Microcenter	Supplier	Microcenter	(800) 207-3434
Amazon	Supplier	Amazon	1-888-280-4331
Customers	End Users	Public	n/a
Website Host	Platform	GoDaddy	1-480-366-3549
Fedex	Shipping	Fedex	1-800-463-3339

Project Manager

Name the selected project manager and provide his or her contact information.

Vitaliy Gotsko

vg534@njit.edu

Project Authorization *(For Professor Use ONLY)*

Date: _____

By initialing each page and signing below, I _____, the Project Sponsor, approve the project described herein and authorize it to begin.

(Insert Name of Organization)

By: _____
Signature of Project Sponsor

_____
Project Sponsor Printed Name